Product Quality 2026: What the Data Shows and a Quality Model

Management update v3 | QA Architecture | 25 October 2026

Bugs got more severe; customers still find them first

+49%

P1/P2 customer-found bugs

Jan-Sep 2026 vs Jan-Sep 2025

+13%

all customer-found bugs

same window; MI investigations removed

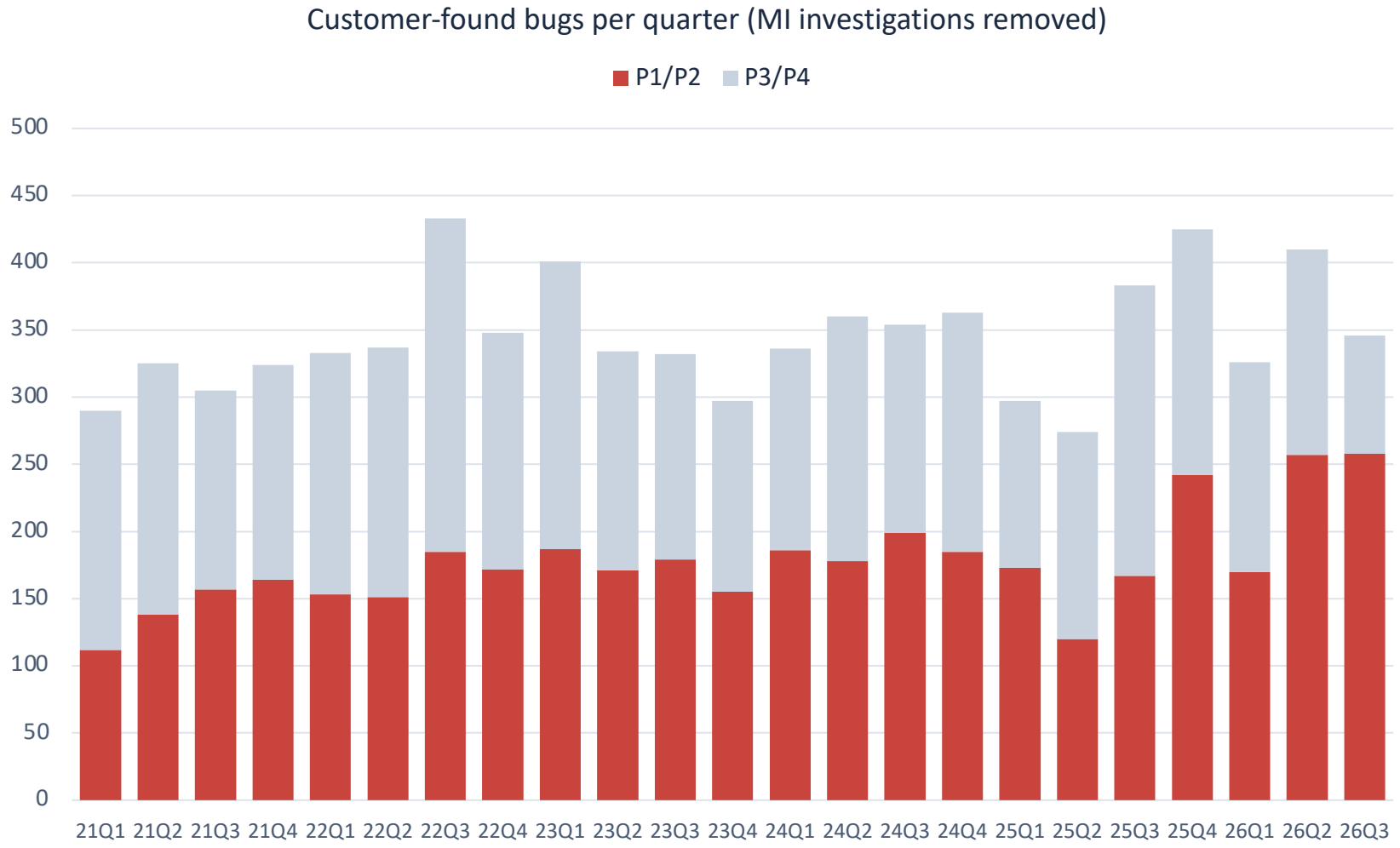
96%

of P1 incidents reach us via customer cases

ServiceNow, since 2025

1. **Severity, not volume:** P1/P2 rose sharply; total volume rose a little and fell in Q3.
2. **Not AI per change:** matched comparisons show no higher bug rate for AI-made changes.
3. **Proposal:** a quality model with a common baseline, risk-based rules, team adaptation, and 5 experiments.

Severity broke the trend in 2026; volume did not



258 vs 167

P1/P2 in Q3 2026 vs Q3 2025

346 vs 383

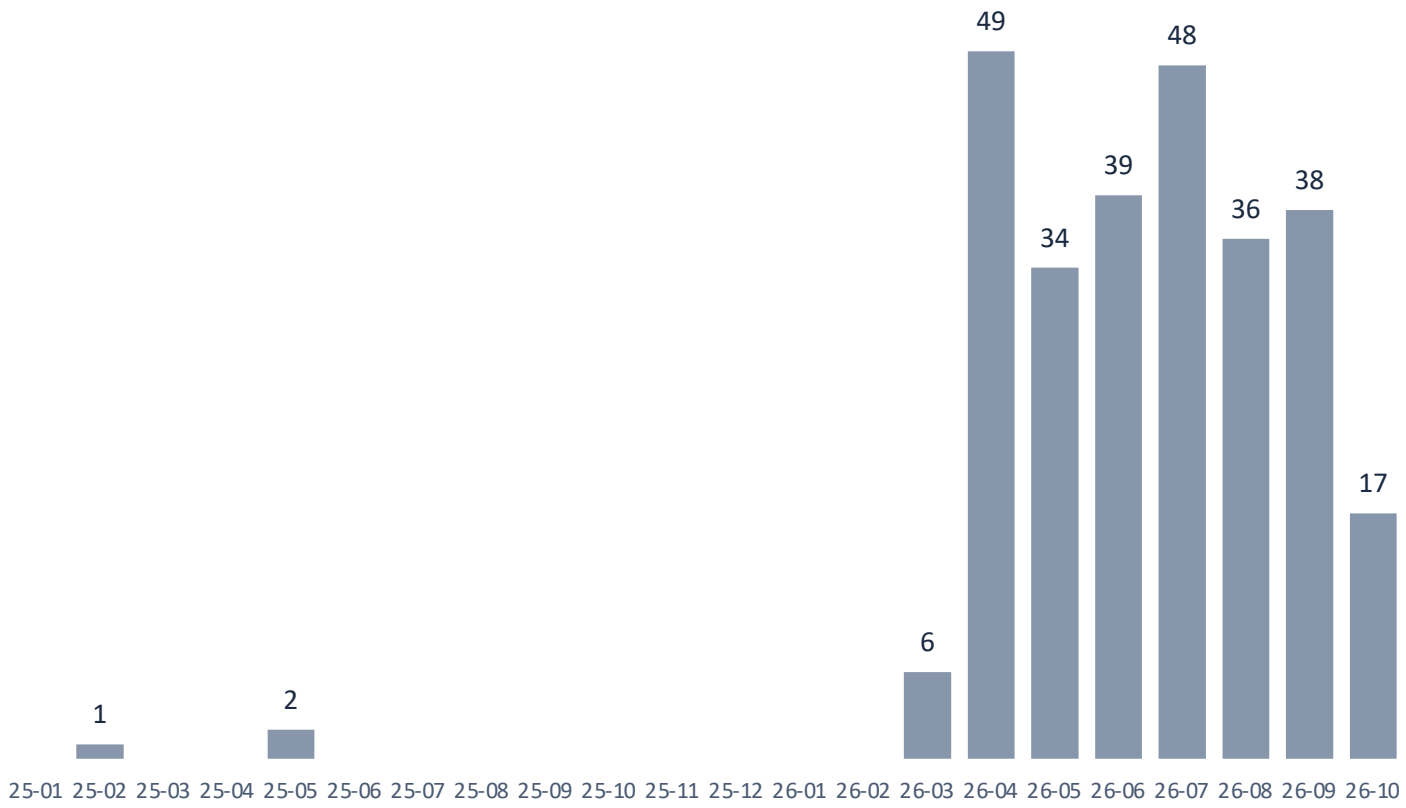
all bugs in Q3 2026 vs Q3 2025

75%

P1/P2 share in Q3 2026 (about 50% in 2023-24)

Correction: our first headline counted MI investigations as bugs

MI investigation tickets logged as customer bugs, per month



Earlier (H1)

+51% bugs, +72% P1/P2

Corrected (Jan-Sep)

+13% bugs, +49% P1/P2

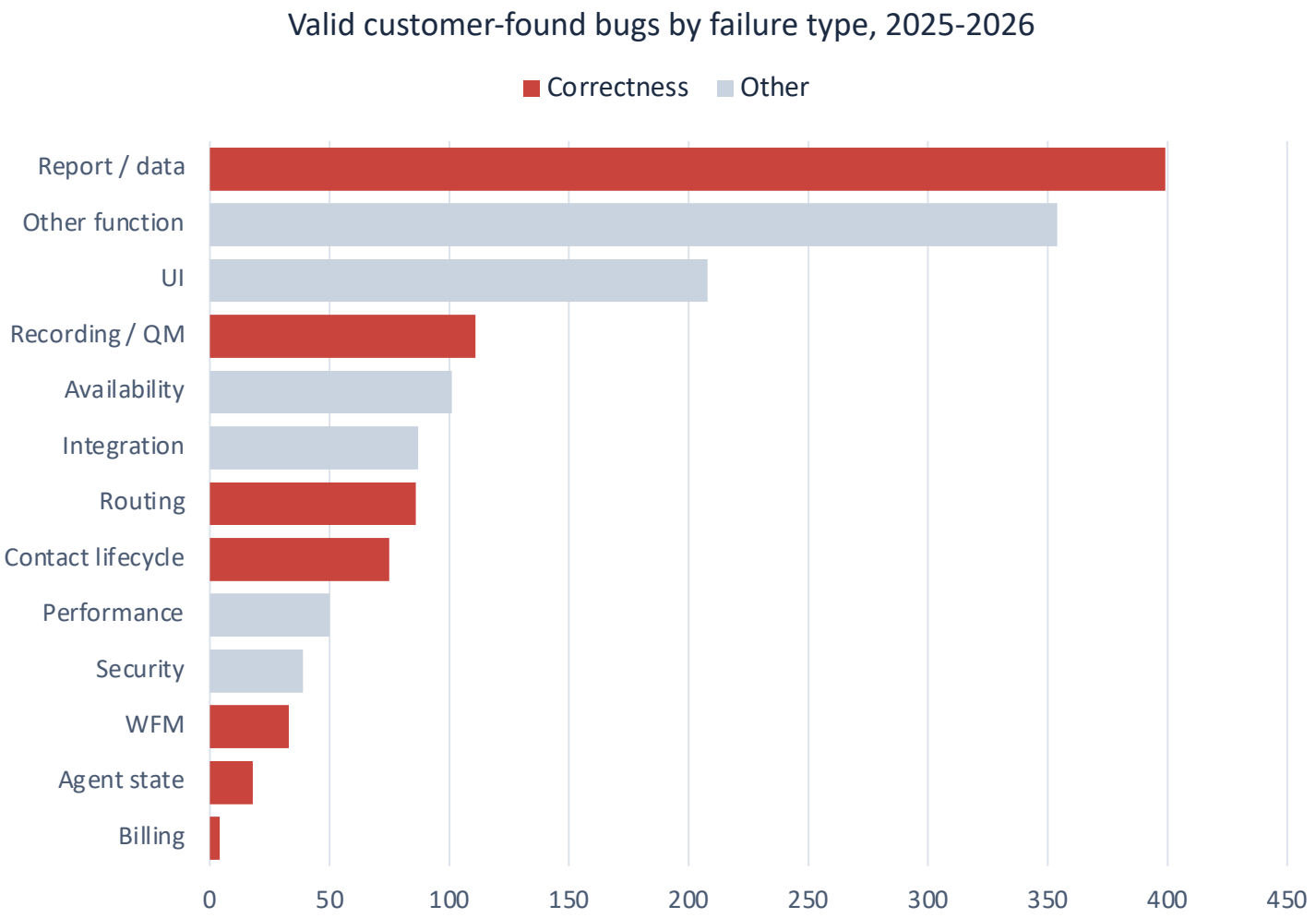
Why

From March 2026, every major-incident investigation is logged as a bug with a customer case

Lesson

We now audit what enters a count before we trend it

Almost half of valid bugs are business-correctness failures



73%

of routing bugs are P1/P2 (highest)

11

incidents per agent-state bug

52% -> 70%

P1/P2 share of correctness bugs, 2025 -> 2026

AI Quality Discovery | Data as of 2026-10-25 | Sources: Jira, Confluence, GitHub, ServiceNow

Teams differ too much for one workflow

1.1 - 10.4

incidents per bug

10 - 93%

P1/P2 share

6 - 79%

invalid bug share

3 - 31

days to fix (median)

Range across 40 teams. The VP explains only 17-46% of this variation; most of it is inside a VP.

High impact

9 teams

Severity and incidents above median

Fast growing

8 teams

Valid bugs +50% or more; part is re-homing

High noise

5 teams

Half or more of bugs invalid

Stable

18 teams

No extra signal

AI: what is fact, what is correlation, what is hypothesis

FACT	AI use grew from about 0% to 31-47% of work in 2026
FACT	AI labels on bugs mark AI-made fixes, not AI-caused bugs
NOT SUPPORTED	AI changes cause more bugs per change (matched ratio 0.78-0.84)
NOT SUPPORTED	AI adoption caused the 2026 rise (adopting teams stayed flat)
WEAK SIGNAL	Same author: AI PRs need more follow-up fixes (1.45, CI touches 1)
MECHANISM	AI PRs are 3.7x larger and 47% of PRs into hotspot code; review does not grow with size

Quality = 8 outcome dimensions, each tied to a decision

Dimension	Decision it supports	2025 -> 2026
Business correctness	Where to invest first	295 -> 325 valid; P1/P2 52% -> 70%
Availability	Reliability, release risk	Counts only (access)
Performance	Capacity work	No SLO data yet
Detection	Monitoring investment	96% of P1 via customer cases
Recovery	Response capacity	P1 71% < 4 h; fix median 10 days
Change safety	Review depth for fixes	1 in 4 causes is a fix; emergency x2.9
Non-recurrence	Do fixes hold	1-2% return; flat
Expectation fit	Requirements and docs	~15% of reports are real gaps

Code quality, coverage and tool scores are inputs that may explain these outcomes. They are not quality.

Common where it adds value, different where risk differs

Common baseline

All teams

- One intake rule: defect vs MI vs expectation gap
- Each bug: type, detected-by, cause, customers
- RCA for P1/P2 (already uniform)
- Team -> service -> code map

Risk-based layer

Any team, by risk

- High-severity correctness failures
- Hotspot code: review scales with size
- Bug fixes: regression evidence
- Emergency changes: follow-up review

Team adaptation

By archetype

- High impact: detection and recovery
- Fast growing: check re-homing, then triage
- High noise: expectation gaps first
- Stable: baseline only

Teams choose how to meet a rule. The model fixes what must be true and how it is measured. Rules apply to all authors, human or AI.

Five experiments, redesigned so the data can show a result

Experiment	Arms	Primary measure	Duration
X1 Intake fields	3 teams (one per type) vs 3 controls	Bugs with cause and detected-by (15% today)	1 quarter
X2 Detection	1 high-impact team, shadow mode	Alerts before a customer case; lead time	6 weeks + 1 quarter
X3 Review by size	All teams with hotspot PRs, 2-step start	Follow-up fixes per hotspot PR (about 70%)	2 quarters
X4 Fix safety	Same pool, bug-fix PRs only	Follow-up fixes per bug-fix PR	2 quarters
X5 Expectation gaps	5 high-noise teams + VMS/QM, WFM	Share of 'works as designed' closures with an action	1 quarter

Power check: bugs are rare per team, so 1 team for 1 quarter cannot show a 30% change for X2-X5 (4 to 33 quarters needed). The new design pools teams and measures events with more volume. Bug counts stay as the slower confirming signal. Each experiment is pre-registered with success and stop rules.

What to keep, change, stop and start

Keep

- RCA for P1/P2 bugs (80-100% today)
- Feature toggles (no monolith rollback)
- AI development, same rules as all changes
- Xray test management

Change

- Intake: defect vs MI vs expectation gap
- Go/No-Go: from "TBD" to data
- E2E: trusted signal before blocking
- Review depth scales with size in hotspots
- QSU: valid, severity-weighted, per team

Stop

- Raw bug count as the quality number
- One gate set for all teams
- Unused Jira quality fields (9 of 15)
- "Mandatory" with no control behind it

Start

- 8-dimension quality model
- Team -> service -> code map
- Monitors for routing, lifecycle, data
- Expectation-gap triage; owner backups
- Experiments X1-X5

Our gates and quality automation: means, not goals

Means	Recommendation	When
Gates (G1-G6)	Few blocking checks where risk is proven: intake fields, hotspot size and review	Now
Go/No-Go	Data-based: 8 dimensions, open P1-P3, E2E health	Now
E2E automation	Repair trust first; then block on suites that hold their pass level	Now -> Q1
Sonar, dependency scan	Keep as build checks	Keep
RIA agents	Later, aimed at fixes and hotspots (RIA misses = 7%)	H2 2027
Mutation testing	Later, only if X3/X4 show weak tests (weak tests = 3%)	H2 2027
Agent harness	Build only for checks that experiments prove	H2 2027

Blocking is justified only where data proves the risk. Everything else waits for the experiment results.

When: measure first, test in a few teams, then spread

Now

Nov 2026

- Decisions: sponsor, X1-X5, ServiceNow access, team map
- Start X1 intake and X2 detection (shadow)
- QSU: valid, severity-weighted, per team
- Triage 13 failing routing E2E scenarios

Q1 2027

Jan-Mar

- X3/X4 step 1 (hotspot and bug-fix PRs)
- X5 with product owners
- Monthly quality review per team
- Re-test the AI effect (90 days)

Q2 2027

April review

- X3/X4 step 2
- Results vs controls
- Model v1 with justified targets
- Decide which checks block

H2 2027

Jul-Dec

- Spread proven practices to all domains
- Decide RIA agents, mutation, harness from evidence

What we need from management

1. **Approve experiments X1-X5 in volunteer teams**

   1 quarter; no org-wide change yet

2. **Grant ServiceNow record-level read for research**

   Needed for who was affected and change-to-incident links

3. **Name an owner for the team-to-service-to-code map**

   No such map exists today

4. **Name a sponsor and a monthly quality review**

   Each team vs its own baseline, 8 dimensions

Timeline

Now	Start X1-X5
2027-Q1	Re-test the AI effect (90 days)
2027-04	Review experiments vs controls
Then	Spread what worked; model v1